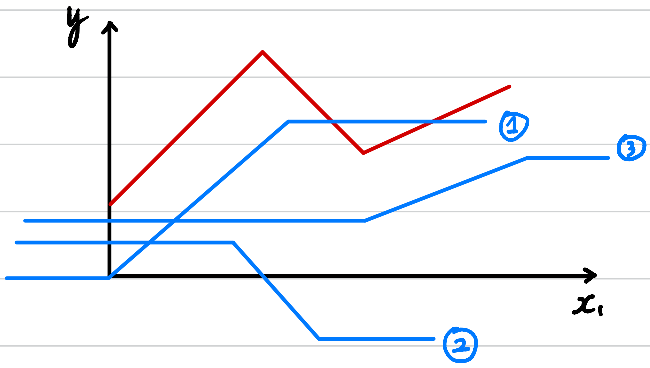
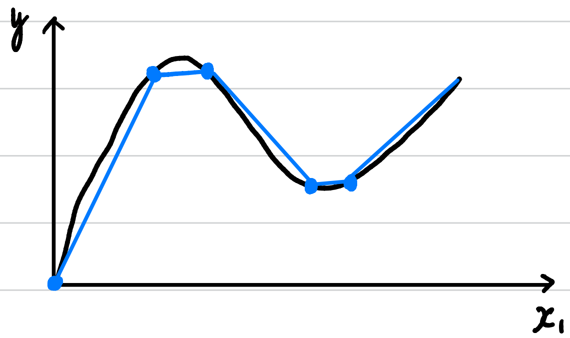
机器学习—台大李宏毅

Episode 2 **线性回归 liner regression**



red curve 可看做 = Constant + a set of _/‾

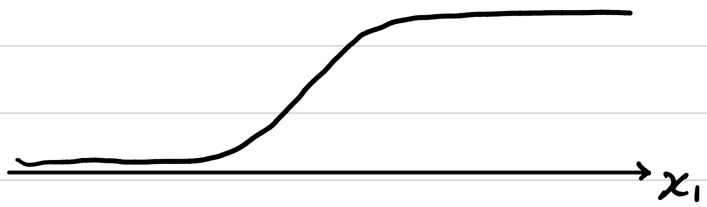
Piecewise(分段函数)可由上面的公式拼成
如左图所示



甚至是曲线也可以由一系列 _/‾ 近似拼凑而成

在此特别介绍 sigmoid 函数: $y = c\,\frac{1}{1+e^{-(b+wx_1)}}$

$= c \text{ sigmoid}(b+wx_1)$



既然如此 $y = b + wx_1$ (红色线) 可用不同的 pairwise liner curves 去逼近

$\Downarrow$

$y = b + \sum_i c_i \text{ sigmoid}(b + w_i x_1)$

那么逼近后如何减少 bias 呢?

举个例子来说：用多个参数来估计Youtube观看人数

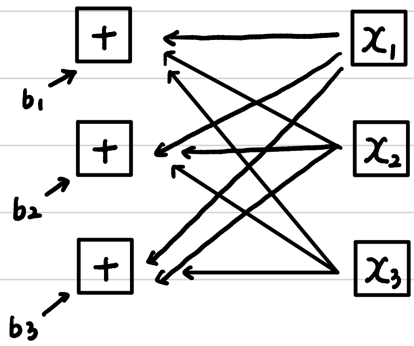
$$y = b + \sum_i c_i \ sigmoid\left(b_i + \sum_j w_{ij} x_i\right)$$

假设 $i=3$ 有3个 sigmoid；$j=3$ 有3个 features

$$r_1 = b_1 + w_{11}x_1 + w_{12}x_2 + w_{13}x_3$$

$$r_2 = b_2 + w_{21}x_1 + w_{22}x_2 + w_{23}x_3$$

$$r_3 = b_3 + w_{31}x_1 + w_{32}x_2 + w_{33}x_3$$



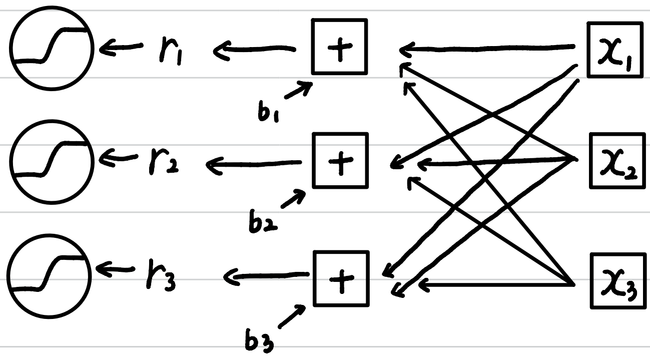
$$\begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} + \begin{pmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\boxed{r} = \boxed{b} + \boxed{W} \quad \boxed{x} \qquad (\mathrm{I})$$

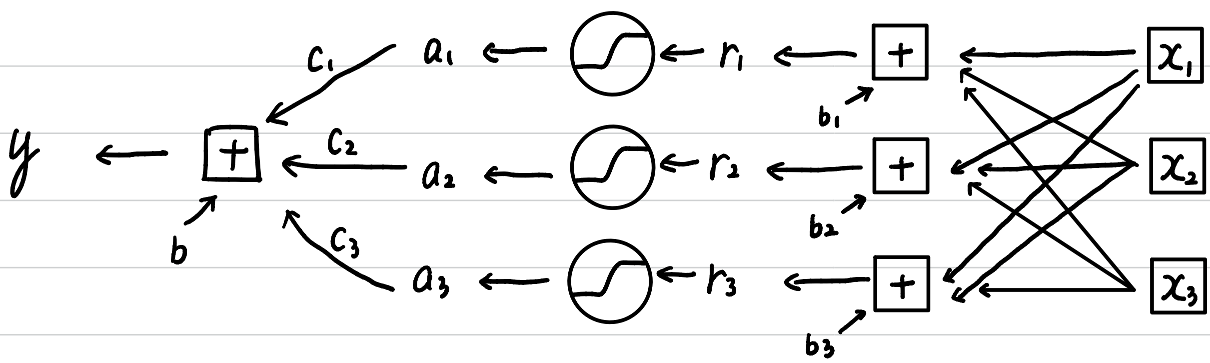
$$a_1 = sigmoid(r_1)$$

$$a_2 = sigmoid(r_2)$$

$$a_3 = sigmoid(r_3)$$



$$\boxed{a} = \sigma(\boxed{r}) \qquad (\mathrm{II})$$

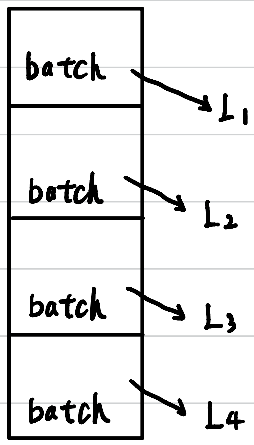


$$y = \boxed{b} + \boxed{c^T}\boxed{a} \qquad (\mathrm{III})$$

$y = \boxed{b} + \boxed{c^T}\,\sigma(\boxed{b} + \boxed{W}x)$ 代入(Ⅰ, Ⅱ, Ⅲ)

其中x是feature，unknown parameters是 $\boxed{W}$, $\boxed{b}$, $\boxed{c^T}$, $\boxed{b}$

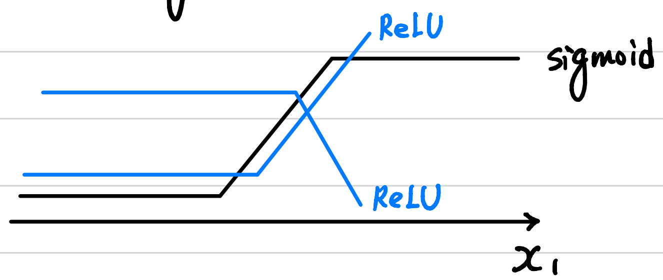
把它们组成一个matrix θ，我们现在想办法优化θ，找到一组θ，让Loss越小越好 同上一节 不断用Gradient Descent $\theta^n \leftarrow \theta^{n-1} - \eta \frac{\partial L}{\partial \theta}\Big|_{\theta=\theta^{n-1}}$



每一个batch中计算一个梯度
之后在下一个batch中update
算完一轮叫一个epoch

模型变型

Sigmoid 可以更换成其它 function like Rectified Linear Unit (ReLU)



Sigmoid可看做由2个ReLU拼凑而成

ReLU $c * \max(0, b + wx_1)$

那么用ReLU替换sigmoid在预测效果上会好一些